



The influence of environmental and physiological factors on the litter size of wild boar (*Sus scrofa*) in an agriculture dominated area in Germany



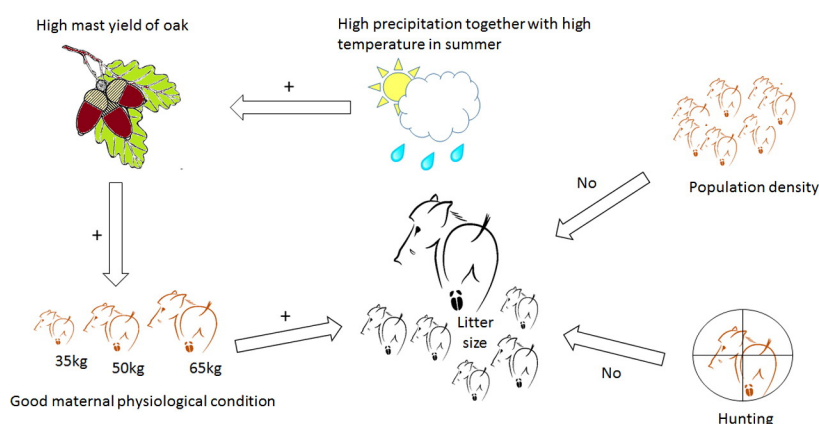
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HIGHLIGHTS

- Litter size of wild boar is positively influenced by higher maternal body weight
- High precipitation combined with high temperature in summer increased litter size
- Higher mast yield of oak also had a positive effect on the litter size
- Climate and nutrition might influence the litter size acting through the body weight
- Population density as well as hunting did not affect the litter size production

GRAPHICAL ABSTRACT



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ABSTRACT

The wild boar population has increased enormously in all of Europe over the last decades and caused problems like crop damage, transmission of diseases, and vehicle accidents. Therefore, it is necessary to investigate the underlying causes of this increase in order to be able to manage populations effectively. The purpose of this study was to analyse how environmental (food and climate) and physiological factors (maternal weight and age) as well as hunting and population density influence the litter size of wild boar populations in Northern Germany. The mean litter size in the studied population for the whole period was 6.6 (range 1–12), which is one of the highest in all of Europe. Litter size was positively influenced by maternal body weight, higher mast yield of oak as well as higher temperature in combination with higher precipitation in summer. Only higher temperature or only higher precipitation in summer however had a negative effect on litter size production. Probably, weather and food conditions act via maternal body weight on the litter size variation in wild boar. Hunting as well as population density did not affect the litter size variation in this study which might indicate that wild boar population did not reach carrying capacity yet.

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1. Introduction

Most ungulates are thought to be typical K-strategists characterized among others by a relative low litter size (Asher, 2011; English et al., 2012).

The wild boar however, is a unique species showing traits from both K- and r-strategists (Gaillard et al., 1989). It shows sexual size dimorphism (Focardi et al., 2008) and a low natural adult mortality rate in temperate climates (Keuling et al., 2013a), which is typical for K-strategists. On the other hand, it has by far the highest reproductive potential and fecundity of all ungulate species world-wide in relation to its body mass (Boitani et al., 1995b; Focardi et al., 2008; Fonseca et al., 2011; Gaillard et al., 1993; Servanty et al., 2009), so that it can reach high population densities within a very short time period (Geisser and Reyer, 2005). Additionally, it is also an opportunistic omnivore with a high adaptability to changing environmental conditions (Baubet et al., 2004; Gamelon et al., 2013; Schley and Roper, 2003; Servanty et al., 2011).

This possibility of varying between K- and r-strategist features depending on environmental conditions has caused an enormous increase of wild boar in the whole of Europe in the last decades (e.g. Keuling et al., 2013a; Lombardini et al., 2014; Massei et al., 2015). Therefore, it has become a very important species from an economical point of view (Kramer-Schadt et al., 2007; Uzal and Nores, 2004). It can act as a reservoir of disease that can be transmitted to livestock (Gortázar et al., 2007). This problem becomes more urgent under high population densities (Cellina, 2008). The spread of the African swine fever (ASF) in Europe among the dense wild boar populations is a recent risk (Sánchez-Vizcaíno et al., 2013) for animal welfare and economy (Mur et al., 2012). In agricultural dominated areas, wild boar cause enormous damage in crop fields (Geisser and Reyer, 2005; Herrero et al., 2006; Keuling et al., 2014).

Therefore, it is necessary to investigate the underlying causes of the increase of this ungulate, to identify appropriate management actions to reduce its density (Keuling et al., 2008; Sodeikat and Pohlmeier, 2007; Uzal and Nores, 2004).

Factors determining various aspects of the reproductive biology of wild boar are still the main area of research (Fernández-Llario et al., 2003) because it directly affects population dynamics (Maillard and Fournier, 2004). The litter size of wild boar is higher than that of other similar-sized ungulates (Carranza, 1996) and, even in comparison with several mammal species, wild boar belongs to the species with the highest litter size (Millar and Zammuto, 1983). Therefore, it is an important parameter in the population growth and life history of wild boar (Fernández-Llario and Mateos-Quesada, 1998). In general, different selective pressures that are caused by changes in food resources, climate, predation or competition shape demographic tactics in animal populations (Servanty et al., 2011). Thus, the purpose of this study was to analyse how environmental (food and climate) and physiological factors as well as hunting and population density influence the litter size of wild boar populations in Germany.

Several studies found a positive relationship between female age or body weight and the number of fetuses per female (Aumaitre et al., 1982; Fernández-Llario et al., 1999; Fonseca et al., 2011). Also environmental factors (weather conditions and food resources) influenced the reproduction of wild boar (Bieber and Ruf, 2005; Fernández-Llario and Mateos-Quesada, 2005; Geisser and Reyer, 2005). Hunting and population density might have an effect on the litter size production as well (Massei et al., 1996; Servanty et al., 2011; Vetter et al., 2015). Due to their large size and long life span, normally, ungulate populations are close to carrying capacity (McCullough, 1999). However, if population density cycles, females produce fewer high quality offspring at peak density (Sinervo et al., 2000) so that we can expect a lower litter size when wild boar density is high (Massei et al., 1996).

Even if some studies about environmental effects (Bieber and Ruf, 2005; Fernández-Llario and Mateos-Quesada, 2005; Geisser and Reyer,

2005; Servanty et al., 2009), physiological effects (Fernández-Llario et al., 1999; Fonseca et al., 2011; Sáez-Royuela and Tellería, 1987) and hunting on the reproduction of wild boar have already been published in Europe, there are still contrasting results on several aspects as for instance concerning the effect of hunting on wild boar populations (Gamelon et al., 2011; Servanty et al., 2011; Vetter et al., 2015).

Thus, our hypotheses in this study are that higher body weight, favourable weather conditions and higher food supply increase the litter size in wild boar. Reduced litter size will be found at high population density whereas high litter size will occur under high hunting impact.

2. Material and methods

2.1. Study area

Our study area was situated in the east of the federal state of Lower Saxony (Northern Germany) (52.36°N, 10.35°E) (Supplementary material I) where altitude ranged from 60 to 130 m a.s.l. (Gethöffer, 2005). It was affected climatically by the maritime climate in the northwest, with low variations in temperature during the course of the year, and the continental climate in the southeast, with greater temperature variation between summer and winter. Based on data from the German Meteorological Service, the average annual temperature from 2000 until 2010 was 9.7 °C (DWD, 2015). The average annual precipitation per year was 807 mm from 2000 until 2010 (DWD, 2015).

The size of the study area was about 4885 km². Cultivated fields and forest comprised 54.6% and 27.6% of the area, respectively. The remaining 17.8% were buildings and infrastructure (NLS, 2007).

The agricultural land was characterized by 43.3% grain (oat, barley, rye and wheat), 16.0% grassland, 11.7% sugar beets, 6.6% oilseed rape, 6.0% maize, 5.8% potatoes. The remaining 10.6% consisted of heathland, fallow land and horticultural products (NLS, 2007).

The forest consisted of 18.9% oak (*Quercus* spp.), 11.2% beech (*Fagus sylvatica*), 7.6% other deciduous trees (e.g. *Betula* spp., *Acer* spp., *Tilia* spp., *Sorbus aucuparia*), 45.5% pine (*Pinus sylvestris* and *Pinus strobus*) and 16.6% other coniferous trees (*Larix* spp., *Pseudotsuga menziesii*, *Picea abies*).

Supplementary feeding was not allowed in the study area. However baiting in the winter with the help of maize is a necessary hunting strategy (Keuling, 2015). Not more than 1000 kg supplemental food per 100 ha was supplied every year for baiting purpose (similar to Keuling et al., 2008).

2.2. Data collection

In total 453 reproductive tracts (uteri and ovaries) were collected from pregnant free ranging hunted wild boar sows in the study area between 2003 and 2014. Wild boar individuals were sampled in the hunting season from October until February (Supplementary material II). Hunting season 2009, for instance, started in October 2009 and went until February 2010. Each individual sampled was associated with one of the nine administrative district (Supplementary material III). Dressed weight (without digestive system, heart, lungs, liver, reproductive tract and blood) of female wild boar was taken. The age of the wild boar sows was identified by tooth determination (Gethöffer et al., 2007). The lower jaw of the sampled individuals was examined, and the tooth eruption, replacement, and dental abrasion were assessed. Based on this age identification, wild boar sows were pooled in three classes (piglets: <13 months, yearlings: 13–24 months, adults: >24 months) (Supplementary material II). During dissection in the laboratory, the number of fetuses was counted.

Meteorological data (temperature and precipitation per month) were supplied by local meteorological services (DWD, 2015). The mean temperature and precipitation of the two seasons, summer (June, July, August) and autumn (September, October, November), before/during data collection was calculated. Each sampled individual

was associated with an average temperature and precipitation per month in summer and autumn, depending on the hunting year and administrative district where the individual sample was collected.

Summer climate data indicated dry summers in 2003, 2009 and 2013, whereas the summers of 2004, 2007 and 2011 were relatively wet compared to the other years during the study period (Supplementary material IV). The summers were relatively warm in 2003, 2006 and 2010 and cold in 2005 and 2012 compared to the other years during the study period (Supplementary material IV).

In 2006, 2011 and 2014, mean autumn precipitation was low whereas it was high in 2007, 2009 and 2010 (Supplementary material V). The years 2003, 2007 and 2010 had a cold autumn whereas in 2006 the mean autumn temperature was quite high compared to the other years of the study period (Supplementary material V).

Data on the amounts of acorn and beechnut crops were delivered by The Forestry Seeds Advisory Board (FSB) at Oerrel/Lower Saxony, Germany. The yield values of beech (*F. sylvatica*) and oak (*Quercus robur* and *Quercus petraea*) per year (from 1 July until 30 June) for Lower Saxony were used for data analysis to identify the effect of food abundance on the litter size.

Each sampled individual was also associated with a hunting index (HI) of the administrative district where it was collected (Supplementary material I). The HI was defined as the hunting bag per km². The hunting bag depends on the way of hunting (Keuling et al., 2013b). Therefore samples were only taken from wild boar hunted by driven hunts, which are less selective compared to single hunts (Keuling et al., 2013b), and there were no limitations in sex and age (Keuling, pers. comm.). This represents the true population structure in a better way because it reflects the hunting coincidence (Keuling et al., 2013b) and it is expected that the hunting effort is the same in each year.

2002 and 2008 were years of high hunting index (Supplementary material VI). The HI of the previous year compared to the year of data collection was used to indicate the possible effect of hunting on the litter size.

To be able to analyse the effect of wild boar density on the litter size, estimations of the abundance of wild boar in the study area are necessary. Unfortunately, the expense and effort required to estimate true population densities are prohibitive (Acevedo et al., 2007). Data required by trap cam method (Rowcliffe et al., 2008) were only partly available for the study area so that the HI was used as an indication for the wild boar population density (Acevedo et al., 2007; Boitani et al., 1995a; Fernández-Llario et al., 2003). Similar to the approach of Vetter et al. (2015), the HI from the same year compared to the year of data collection was used to identify a possible effect of population density on the litter size, knowing that this is only a proxy of the density and should be regarded with caution.

Similarly, the HI of the following year compared to the year of data collection was used to get an impression of the response of hunters on changing reproductive effort of wild boar.

2.3. Statistical analysis

For all tests, significance was assumed when $p < 0.05$. All analyses were performed using R software version 3.1.2 (R Core Team, 2014). Generalized linear models (GLM) were performed. Since the dependent variable, litter size, was a count variable ranging from 1 until 12, Poisson distribution was used. Similar to the statistical approach of Servanty et al. (2009), the analysis was broken down in two steps to avoid overparametrization in the models. First effects of body weight and food condition were tested together with their possible interactions, subsequently, the additional effects of weather inclusive possible interactions within weather conditions were tested. Because maternal body weight and age class correlated ($p = 0.79$) only maternal body weight was included in the model. Data on body weight might be more reliable as errors could occur during determination of the age classes with the help of tooth eruption depending on different observers.

Secondly, a generalized linear model was performed to identify the effect of hunting and population density on the litter size as well as the response of hunters on the litter size. The interaction between those variables was also tested. This model was conducted separately because concerning the HI data from one year (2014) were missing ($n = 395$) compared to the remaining data of weather and food conditions ($n = 453$).

Explanatory variables were checked for correlations between each other. If a correlation was larger than 0.7, one of the two variables was excluded from analysis (Zuur et al., 2010). Models were validated by checking cooks distance and temporal autocorrelation.

Pearson dispersion statistic was under 1 indicating underdispersion (Hilbe, 2014). VGAM package in R was used to correct for underdispersed data (Yee, 2010). Correcting for underdispersion was only possible with the VGAM package, however, using mixed models is not possible at the same time. Therefore, a choice had to be made. Ignoring underdispersion resulted in different outcomes than correcting for underdispersion (within the package VGAM). On the other side, including or excluding administrative district and year as random factor (within the package lme4) resulted in same results. This is why the VGAM package was used to correct for underdispersion and no random factors were included in the models.

The Akaike Information Criterion (AIC) (Burnham and Anderson, 2002) was used to drop each explanatory variable in turn until the best fitting model was reached. The model with the lowest AIC was selected as the best one.

3. Results

The mean litter size in the studied population for the whole period was 6.6 (± 2.0 SD; $n = 453$) ranging from 1 to 12 fetuses per sow. The best fitting model (Supplementary material VII) included maternal body weight, as well as food and weather conditions. Litter size increased with higher body weight (Fig. 1). In addition, higher mast crops of oak showed a positive effect on litter size in the studied population (Table 1). Temperature and precipitation in summer had both a significant negative effect on the litter size, whereas those two variable in interaction showed a positive effect on the litter size (Table 1), indicating that high temperature in combination with high precipitation in summer resulted in higher litter size.

No significant effect of mast of beech (GLM, $z = 0.565$, $n = 453$, $p = 0.57$) as well as temperature in autumn (GLM, $z = -0.761$, $n = 453$, $p = 0.45$) or the interactions of temperature in autumn with temperature in summer (GLM, $z = 1.899$, $n = 453$, $p = 0.058$) or precipitation in summer (GLM, $z = 0.284$, $n = 453$, $p = 0.78$) were found.

The best fitting model concerning the HI (Supplementary material VIII) indicated no significant relationship between population density (indicated by the HI of the same year) and litter size (GLM, $z = 0.190$, $n = 395$, $p = 0.849$). Also the HI of the following year compared to the year of data collection showed no relationship with litter size (GLM, $z = 1.524$, $n = 395$, $p = 0.127$). This means that there is no response of hunters on litter size variation in wild boar. In addition, interactions of those two variables did not influenced the litter size (GLM, $z = -0.961$, $n = 395$, $p = 0.336$). Since the HI from the previous year compared to the year of data collection was already excluded during model selection process, we can conclude that also hunting did not show any significant effect on the litter size variation in wild boar.

4. Discussion

Results show that litter size was influenced by physical maternal condition (body weight). Also environmental parameters affected wild boar litter size. Better mast crop conditions of oak resulted in a higher litter size of wild boar sows. Higher precipitation and temperature in summer led to decreased litter size. On the other hand, high temperature in combination with high precipitation in summer led to increased

litter size. However, litter size was neither affected by hunting nor by population density. In addition, no response of hunters on litter size variation was found.

A mean litter size of 6.6 fetuses per female wild boar is recorded to be one of the highest in Europe (Fonseca et al., 2011).

In Southern Europe (Fernández-Llario and Mateos-Quesada, 2005; Lombardini et al., 2014; Rosell et al., 2012; Servanty et al., 2007), in Northern Europe (unpublished, Malmsten et al.) as well as in mountainous regions litter sizes are lower (Moretti, 1995).

Our results are supported by several studies where a positive correlation between body weight and litter size was found (Boitani et al., 1995b; Fernández-Llario et al., 1999; Fernández-Llario and Mateos-Quesada, 1998; Fonseca et al., 2011; Rosell et al., 2012). During the 12-year study period, average dressed maternal body weight increased in piglets, yearlings and adults with 12%, 13% and 11%, respectively (Lustig, 2015). This should be seen as a sign of caution that litter size might increase in the future when weight increases if the current climatic trend goes on. This statement is supported by the results documented by Appelius (1995) who found an average litter size of 4.67 and 5.11 for piglets and adults, respectively, for the same study area from 1993 until 1995. Our results are in line with the outcomes found in Fonseca et al. (2011) and Rosell et al. (2012) where maternal weight had a positive influence on the litter size. A possible explanation for the correlation found between maternal weight and age could be that adults have a significantly higher weight than piglets so that they can invest more energy in a higher litter size.

Both temperature and humidity affects the reproductive physiology and population dynamics in the terrestrial ecosystem (Walther et al., 2002). An increase in temperature caused by global warming in the

last decades has led to a higher frequency of mast years in beech (Övergaard et al., 2007; Vetter et al., 2015). As oak and beech are the most important natural sources of food for wild boar (J drzejewska et al., 1994), it results in large amounts of food (J drzejewska et al., 1997) and higher maternal body weight that increases the energy to invest in offspring (Bieber and Ruf, 2005; Geisser and Reyer, 2005; Melis et al., 2006). This might be the reason for the population increase in the whole of Europe. Acorns are high in carbohydrates, which are essential for reproduction and maintenance of good physical condition (Massei et al., 1996). This can explain our findings about a significantly higher litter size in years of high acorn supply. Several studies found a correlation between good mast years and a large litter size as well (Fernández-Llario et al., 1999; Geisser and Reyer, 2005). Litter size was not affected significantly by beech mast in contrast to the study from Vetter et al. (2015) who found a significant effect of beech mast on the population growth in wild boar. A possible explanation is the lower presence of beech compared to oak in the study area. This could make effects less obvious compared to oak yields. Interactions of oaks and beech mast conditions were not significant and therefore not mentioned.

Supplementary feeding by baiting might also have an impact, but in comparison with the food supply by mast years and agriculture, the amount of maize supplied in winter is minimal (Keuling, 2015).

Since the study area consisted of about 55% agriculture and intensification of agriculture is recognised as one of the most significant human alterations to the global environment (Matson et al., 1997), possible influence of agriculture should be taken into account. Although Schley and Roper (2003) argued that mast is preferred above agricultural crop and supplementary food, an additional stable food supply provided by crop fields and supplementary feeding on game feeding places may buffer the negative impact of bad mast years and help to keep a high reproduction (Bieber and Ruf, 2005; Cellina, 2008; Rosell et al., 2012). Merta et al. (2014) compared diet, carcass weight, and kidney fat of wild boar in two different areas: a forest-farmland environment and a large compact forest area. Kidney fat and weight of piglets and older animals were higher in the forest-farmland environment. Also Geisser and Reyer (2005) found a higher population density of wild boar with greater maize cultivation as well as Neet (1995) showed in a modelling approach that over a 32 years period, the expansion of maize culture explained 49.7% of the variance in the numbers of wild boar.

Groot Bruinderink and Hazebroek (1994) found a relation between weight and mast availability that could indicate an indirect effect of mast via maternal body mass on the litter size. We cannot support the results that different age classes or weight (in our case) responded in a different way to mast availability, as interactions between maternal body weight and food availability were not significant. Those different outcomes can be explained by the fact that wild boar in the Netherlands lived under different conditions, namely without access to agricultural fields.

Hot and rainy summers indicate a longer vegetative growing season (Fenner, 1998), which is probably favourable for wild boar reproduction due to higher food supply. In temperate climates, warm temperatures often act as flowering triggers, especially in mast seeding trees (Fenner, 1998). Probably, higher temperature in summer is favourable for beech and oak, which has an effect on the litter size acting via nutrition. Since the study area consists of 55% agriculture and cultivated landscapes are used by wild boar as additional food source (Keuling et al., 2009), higher temperature and rain in summer might also be favourable for growing of crops. Geisser and Reyer (2005) did not find that precipitation in April until June or snow in December until February had any effect on the population density. The results of this study are in line with the outcome of the research of Fernández-Llario and Mateos-Quesada (2005). They found in Western Spain a significant increase in litter size in females, which were older than 2 years, in rainy years compared to dry years. However, since Western Spain is a drier and hotter region than Northern Germany, precipitation could have a much higher

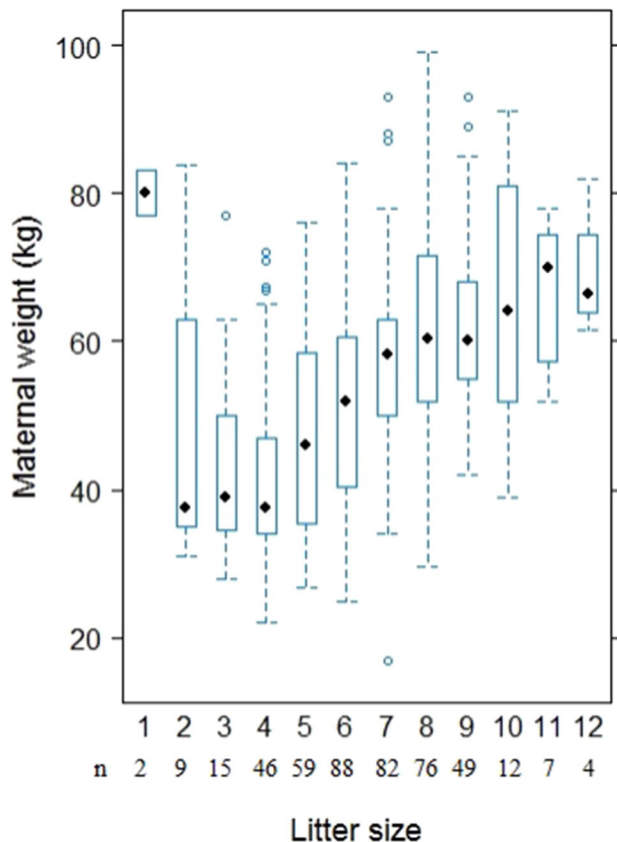


Fig. 1. Maternal body weight in kg per litter size of 12 wild boar hunting seasons in the study area. n indicates the number of samples per litter size. Box and whisker plots show median (point within box), 25% (interquartile range 1) and 75% (interquartile range 3) percentiles (box) and the range (whiskers), which includes observations up to 1.5 times the interquartile ranges. Circles indicate statistical outliers (observations outside 1.5 times the interquartile ranges).

Table 1

GZLM including maternal body weight, food and weather conditions (see [Material and methods](#)) of the 12 wild boar hunting years in the study area ($n = 453$). Precipitation and temperature are for the summer season proceeding data collection. Estimates, standard error (SE) and z-values of the significant effects of the best fitting model are shown. Significance is indicated in the following way: * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$.

Variable	Estimate	SE	z value	Significance	Effect
Body weight	5.656^{-3}	4.614^{-4}	12.258	***	Increased litter size with higher maternal body weight
Mast crops oak	2.858^{-6}	6.896^{-7}	4.145	***	Increased litter size with higher mast crop of oak
Temperature in summer	-4.010^{-2}	1.149^{-2}	-3.490	***	Decreased litter size with higher temperature in summer
Precipitation in summer	-2.216^{-2}	1.076^{-2}	-2.061	*	Decreased litter size with higher precipitation in summer
Temperature in summer * precipitation in summer	2.518^{-3}	7.872^{-4}	3.199	**	Increased litter size with higher temperature in combination with higher precipitation in summer

impact on the litter size. In addition, also [Servanty et al. \(2009\)](#) found a peak in breeding females when temperature and rainfall was high in summer. In contrast to the research from [Vetter et al. \(2015\)](#) we did not find any significant effect of temperature in autumn on the litter size. Those different results might come from the fact that they used the variable population growth as dependent variable. However, their results support our outcome concerning the temperature in summer having a significant negative effect on the litter size.

Nevertheless, the weather conditions seem to have an effect on the litter size of wild boar acting through their nutritional condition.

Against our expectations and in contrast to results from [Vetter et al. \(2015\)](#), no effect of population density on litter size could be found. A possible explanation might be that the hunting index is not reliable enough as indication for the population density. Based on [Acevedo et al. \(2007\)](#), [Merta et al. \(2015\)](#) and [Vetter et al. \(2015\)](#) we assumed that the age structure and density of the culled wild boar sows are representative for the age structure and density of the studied wild boar population. Another reason might be that the population density is on a moderate level (far away from carrying capacity) so that density-dependent changes in life-history traits that occur at population levels close to carrying capacity ([Zedrosser et al., 2006](#)) does not act here.

In this research, a higher HI in the previous year did not influence the litter size significantly indicating that hunting had no effect on the litter size production in the studied population. Those results do not fit with the outcomes documented by [Servanty et al. \(2011\)](#). They showed that female wild boar in a heavily hunted area had a higher generation time than females in a lightly hunted area. In addition, the litter size did not show any significant relationship with the HI of the following hunting year, which means that there is no response of hunters on litter size variation. This could be an indication that hunting is insufficient to limit wild boar population growth ([Csányi, 1995](#); [Massei et al., 2015](#); [Nores et al., 2008](#)). Another possible explanation for this non-correlation could be that the population density of the following hunting season in fact is not higher. Data were collected only until February (Supplementary material II), so that population reduction by perinatal or post-natal mortality could not be recognised in this study.

5. Conclusion

To summarize, we can say that the wild boar litter size production responds to several environmental conditions like mast yield of oak and summer weather conditions, probably acting through the maternal body weight. If the current climatic trend goes on, weather conditions are expected to be advantageous for wild boar in the future. On the other hand, hunting and population density did not influence the litter size in this studied wild boar population which might come from the fact that wild boar population did not reach carrying capacity yet.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.scitotenv.2015.09.128>.

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